

MHD-Guided Scaling and Initial Experimental Evaluation of the High-Density Argon Hall Thruster RAIJIN30

Nadine Barth¹, Kimiya Komurasaki¹, Dibyesh Satpathy¹, Jiwon Lee¹, Maho Matsukura¹, Hiroyuki Koizumi¹

¹ Department of Aeronautics and Astronautics, Graduate School of Engineering, The University of Tokyo.

1. INTRODUCTION

Argon is an attractive alternative propellant for Hall thrusters because of its low cost and high natural abundance. However, its lower ionization cross section and higher ionization energy generally result in lower propellant utilization efficiency than xenon. Increasing the plasma density can partially compensate for these limitations by enhancing the ionization rate within the discharge channel [1,2].

One approach to increasing the plasma density is to reduce the channel cross-sectional area while maintaining a comparable propellant mass flow rate. Conventional Hall thruster scaling laws predict that the magnetic flux density should increase as the thruster dimensions decrease. However, the applicability of these scaling relations to high-density operation remains unclear. This work investigates the feasibility of high-density Hall thruster operation through geometric scaling and presents initial experimental results obtained with the newly developed RAIJIN30 Hall thruster.

2. THRUSTER DESIGN AND SCALING

RAIJIN30 was developed from the existing RAIJIN66 TAL-type Hall thruster [3] by reducing the channel cross-sectional area by a factor of 2.7, thereby increasing the propellant mass flow density by the same factor. The design process was guided by a magnetohydrodynamic (MHD) model relating plasma density, acceleration length, and magnetic confinement. The final geometry represents a compromise between increasing the plasma density and maintaining a practically achievable magnetic field strength.

3. EXPERIMENTAL RESULTS

Initial performance measurements were conducted using argon propellant over a range of discharge voltages from 150 to 300 V, at a constant argon mass flow rate of 2.08 mg/s, and magnetic field strengths from 20 to 60 mT. Figure 1 compares the measured thrust-to-power ratio of both thrusters as a function of specific impulse. Constant anode efficiency contours are included for reference.

For comparable discharge powers, RAIJIN30 produced higher thrust than the baseline thruster. The improved performance is attributed to the higher plasma density resulting from the increased mass flow density, which is expected to enhance the

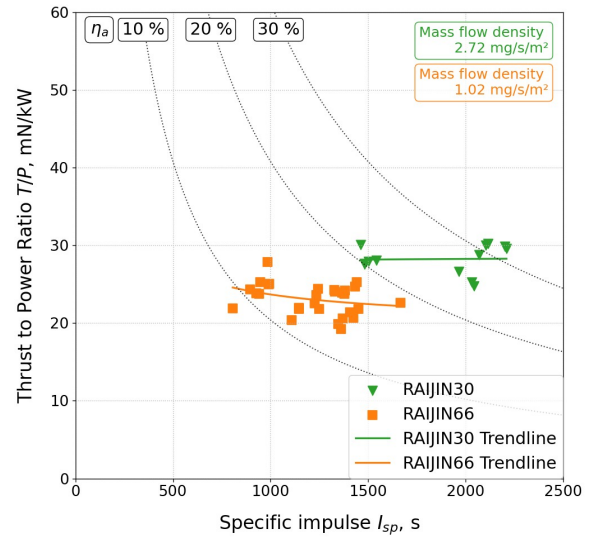


Fig.1 Measured thrust-to-power ratio as a function of specific impulse for both thrusters operated with argon. Solid lines represent hyperbolic fits to the data.

ionization rate and propellant utilization. The largest improvement was observed at higher discharge powers, where the measured thrust increased by 30%. At the same discharge power, the anode efficiency increased by up to 70%.

These initial results indicate that increasing the propellant mass flow density through channel-area reduction can improve the performance of argon Hall thrusters.

4. CONCLUSION

The initial experimental results demonstrate that MHD-guided geometric scaling can improve the performance of argon Hall thrusters by increasing the mass flow density. These findings support further investigation of high-density Hall thrusters using alternative propellants.

REFERENCES

- [1] C. O. Brown, E. A. Pinsley, Further Experimental Investigations of a Cesium Hall-Current Accelerator, *AIAA Journal* 3, 1965, 853-859.
- [2] R.R. Hofer, J. B. Simmonds, D. M. Goebel, J. M. Steinkraus, A. R. Payman, The H10 High power density Hall thruster, *Journal of Electric Propulsion*, 2025, 1-26
- [3] Y. Hamada, R. Kawashima, J. Bak, K. Komurasaki, Characterization of acceleration zone shifting in an anodelayer-type hall thruster RAIJIN66 *Vacuum* 186, 2021, 40.